11 Discrete Mathematics: Set Theory & Algebra (20)

11.1 Abelian Group (2)

11.1.1 Abelian Group: NIELIT 2016 DEC Scientist B (CS) - Section B: 28

Which one of the following is NOT necessarily a property of a Group?

A. Commutativity
B. Associativity
C. Existence of inverse for every element
D. Existence of identity

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Answer key☟

11.1.2 Abelian Group: NIELIT 2017 DEC Scientist B - Section B: 48

On a set $A = \{a, b, c, d\}$ a binary operation $*$ is defined as given in the following table.

$*$	a	b	c	d
a	a	c	b	d
b	c	b	d	a
c	b	d	a	c
d	d	a	c	b

The relation is

A. Commutative but not associative
B. Neither commutative nor associative
C. Both commutative and associative
D. Associative but not commutative

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Answer key☟

11.2 Boolean Algebra (1)

11.2.1 Boolean Algebra: NIELIT 2017 July Scientist B (IT) - Section B: 18

If B is a Boolean algebra, then which of the following is true?

A. B is a finite but not complemented lattice
B. B is a finite, complemented and distributive lattice
C. B is a finite,distributive but not complemented lattice
D. B is not distributive lattice

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Answer key☟

11.3 Cartesian Product (1)

11.3.1 Cartesian Product: NIELIT 2017 July Scientist B (IT) - Section B: 10

What is the Cartesian product of $A = \{1, 2\}$ and $B = \{a, b\}$?

A. $\{(1, a), (1, b), (2, a), (b, b)\}$
B. $\{(1, 1), (2, 2), (a, a), (b, b)\}$
C. $\{(1, a), (2, a), (1, b), (2, b)\}$
D. $\{(1, 1), (a, a), (2, a), (1, b)\}$

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Answer key

11.4 Functions (3)

11.4.1 Functions: NIELIT 2016 DEC Scientist B (CS) - Section B: 34

How many onto (or surjective) functions are there from an n-element $(n >= 2)$ set to a 2-element set?

A. 2^n
B. $2^n - 1$
C. $2^n - 2$
D. $2(2^n - 2)$

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Answer key

11.4.2 Functions: NIELIT 2016 MAR Scientist B - Section B: 8

If $\Delta f(x) = f(x + h) - f(x)$, then a constant k, Δk

A. 1
B. 0
C. $f(k) - f(0)$
D. $f(x + k) - f(x)$

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11.4.3 Functions: NIELIT 2016 MAR Scientist C - Section C: 16

If $f : \{a, b\}^* \to \{a, b\}^*$ be given by $f(n) = ax$ for every value of $n \in \{a, b\}$, then f is

A. one to one not onto
B. one to one and onto
C. not one to one and not onto
D. not one to one and onto

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Answer key

11.5 Group Theory (1)

11.5.1 Group Theory: NIELIT 2016 DEC Scientist B (IT) - Section B: 21

Consider the set $S = \{1, \omega, \omega^2\}$, where ω and ω^2 are cube roots of unity. If $*$ denotes the multiplication operation, the structure $(S, *)$ forms:

A. A group
B. A ring
C. An integral domain
D. A field

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Answer key

11.6 Inclusion Exclusion Principle (1)

11.6.1 Inclusion Exclusion Principle: NIELIT 2017 DEC Scientist B - Section B: 46

The number of integers between 1 and 500(both inclusive) that are divisible by 3 or 5 or 7 is _________.

A. 269 B. 270 C. 271 D. 272

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Answer key

11.7 Lattice (1)

11.7.1 Lattice: NIELIT 2017 July Scientist B (IT) - Section B: 22

Let L be a lattice. Then for every a and b in L which one of the following is correct?

A. $a \vee b = a \wedge b$
B. $a \vee (b \vee c) = (a \vee b) \vee c$
C. $a \vee (b \wedge c) = a$
D. $a \vee (b \vee c) = b$

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Answer key

11.8 Partial Order (2)

11.8.1 Partial Order: NIELIT 2017 July Scientist B (IT) - Section B: 16

A partial ordered relation is transitive, reflexive and

A. antisymmetric B. bisymmetric C. antireflexive D. asymmetric

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Answer key

11.8.2 Partial Order: NIELIT 2017 July Scientist B (IT) - Section B: 17

Let $N = \{1, 2, 3, \dots\}$ be ordered by divisibility, which of the following subset is totally ordered?

A. $(2, 6, 24)$ B. $(3, 5, 15)$ C. $(2, 9, 16)$ D. $(4, 15, 30)$

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Answer key

11.9 Polynomials (1)

11.9.1 Polynomials: NIELIT 2016 MAR Scientist B - Section C: 21

A polynomial $p(x)$ is such that $p(0) = 5,\ p(1) = 4,\ p(2) = 9$ and $p(3) = 20$. The minimum degree it can have is

A. 1 B. 2 C. 3 D. 4

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Answer key☟

11.10 Relations (3)

11.10.1 Relations: NIELIT 2016 DEC Scientist B (IT) - Section B: 28

What is the possible number of reflexive relation on a set of 5 elements?

A. 2^{10} B. 2^{15} C. 2^{20} D. 2^{25}

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Answer key☟

11.10.2 Relations: NIELIT 2017 July Scientist B (IT) - Section B: 21

The relation $\{(1,2),(1,3)(3,1),(1,1),(3,3),(3,2),(1,4),(4,2),(3,4)\}$ is

A. Reflexive B. Transitive C. Symmetric D. Asymmetric

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Answer key☟

11.10.3 Relations: NIELIT 2021 Dec Scientist B - Section B: 115

If A and B are two sets. A binary relation from set A and set B is any subset of the ______________ .

A. Cartesian Product $\mathrm{A} \times \mathrm{B}$
B. Union $\mathrm{A} \cup \mathrm{B}$
C. Intersection $\mathrm{A} \cap \mathrm{B}$
D. Addition $\mathrm{A} + \mathrm{B}$

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Answer key☟

11.11 Set Theory (4)

11.11.1 Set Theory: NIELIT 2016 MAR Scientist B - Section C: 23

I f S be an infinite set and $S_1 \ldots\ldots, S_n$ be sets such that $S_1 \cup S_2 \cup \cdots \cup S_n = S$, then

A. at least one of the set S_i is a finite set.
B. not more than one of the sets S_i can be finite.
C. at least one of the sets S_i is an infinite set.

D. not more than one of the sets S_i can be infinite.

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Answer key

11.11.2 Set Theory: NIELIT 2017 July Scientist B (IT) - Section B: 11

What is the Cardinality of the Power set of the set $\{0,1,2\}$?

A. 8 B. 6 C. 7 D. 9

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Answer key

11.11.3 Set Theory: NIELIT 2017 July Scientist B (IT) - Section B: 20

If A and B are two sets and $A \cup B = A \cap B$ then

A. $A = \phi$ B. $B = \phi$ C. $A \neq B$ D. $A = B$

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Answer key

11.11.4 Set Theory: NIELIT 2017 July Scientist B (IT) - Section B: 9

Power set of empty set has exactly ________ subset

A. One B. Two C. Zero D. Three

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Answer key

Answer Keys

11.1.1	A	11.1.2	A	11.2.1	B	11.3.1	C	11.4.1	C
11.4.2	B	11.4.3	A	11.5.1	A	11.6.1	C	11.7.1	B
11.8.1	A	11.8.2	A	11.9.1	B	11.10.1	C	11.10.2	B
11.10.3	A	11.11.1	C	11.11.2	A	11.11.3	D	11.11.4	B